

Topological Disclination Invariants and Conservative Flux Closures in Icosahedral Hexagonal Discrete Global Grid Systems

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Abstract

Discrete global grid systems (DGGS) built upon icosahedral hexagonal hierarchies, such as the Uber H3 discrete manifold, are bounded by fundamental topological constraints. Under the Euler-Poincaré characteristic for the 2-sphere ($\chi(\mathbb{S}^2) = 2$), uniform hexagonal planar tiling is geometrically impossible; precisely twelve pentagonal faces ($+60^\circ$ Frank disclinations) must exist across any subdivision resolution $r \geq 0$. In dynamic biophysical simulations computing coupled advective-diffusive transport of carbon, water, nutrients, and thermal exergy across spatial dual graphs, numerical conservation laws depend strictly on cell coordination fidelity. Admitting an erroneous sixth neighbor at a pentagonal disclination core introduces non-existent metric boundary facets. These “ghost edges” destroy the skew-symmetry of interface transport tensors ($\Phi_{ij} \neq -\Phi_{ji}$), producing spurious mass-energy generation in violation of the First Law of Thermodynamics and artificial entropy drift violating the Second Law. In Sprint 078 of the Web of Life simulation architecture, we refactor the spatial validation engine to enforce strict coordination invariants. The refactored `assertValidNeighborCountForCell` routine differentiates between topological disclination cores and standard hexagonal faces, explicitly instantiating and raising `PentagonalCoordinationViolationError` whenever a pentagonal cell exhibits a neighbor cardinality $z \neq 5$. Encapsulated within a monadic spatial flux operator (`SpatialFluxMonad`), this check provides fail-fast conservation guarantees prior to state transitions.

1 Introduction

Simulating biosphere-scale thermodynamics on spherical planetary manifolds requires discrete geometric representations that balance equal-area partitioning, minimal angular distortion, and uniform topological coordination [1]. The discrete global grid system (DGGS) based on recursive aperture subdivision of the regular icosahedron—most notably implemented in the H3 architecture [2]—has emerged as a standard computational substrate for geographic continuum mechanics.

In such architectures, standard cells exhibit hexagonal planforms with coordination number $z = 6$. However, by the Euler-Poincaré theorem, a spherical surface cannot be tiled exclusively with hexagons. The curvature of the sphere forces the inclusion of twelve pentagonal topological disclinations at each resolution level [3].

While structural geographic information systems (GIS) frequently tolerate truncated or ambiguous neighborhood queries, numerical continuum mechanics and non-equilibrium ecological thermodynamics demand strict topological consistency. In the **Web of Life** planetary simulator (<https://github.com/pascalranoroarijaona/WebOfLife>), physical state vectors representing carbon, moisture, oxygen, minerals, and thermal energy are transported across directional dual

graph interfaces. If a pentagonal disclination is mistakenly assigned a sixth neighbor—or if neighborhood arrays are truncated without boundary-condition adapters—interface divergence balances break down.

This paper details the mathematical, thermodynamic, and software architectural foundations of Sprint 078, which implements strict coordination validation and introduces the `PentagonalCoordinationViolationError` domain model in `src/spatial/h3_adjacency.ts`.

2 Topological Foundations: Euler Characteristic and Disclination Defects

2.1 The Euler-Poincaré Constraint

Let $\mathcal{M} = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ denote the discrete 2-manifold cell complex representing a tessellated sphere $\mathbb{S}^2$. The Euler characteristic $\chi(\mathcal{M})$ is defined as:

$$\chi(\mathcal{M}) = |\mathcal{V}| - |\mathcal{E}| + |\mathcal{F}| = 2(1 - g) \quad (1)$$

For a 2-sphere with genus $g = 0$, $\chi(\mathbb{S}^2) = 2$. In an icosahedral dual cell decomposition where all vertices are trivalent (each vertex is shared by exactly 3 faces, representing dual three-cell junction centroids), the edge-to-vertex relationship is identically:

$$3|\mathcal{V}| = 2|\mathcal{E}| \implies |\mathcal{V}| = \frac{2}{3}|\mathcal{E}| \quad (2)$$

Let F_k denote the number of k -gonal faces in $\mathcal{F}$. The total face count and the edge count satisfy:

$$|\mathcal{F}| = \sum_{k \geq 3} F_k, \quad 2|\mathcal{E}| = \sum_{k \geq 3} kF_k \quad (3)$$

Substituting Equations (2) and (3) into Euler's formula yields:

$$\frac{2}{3}|\mathcal{E}| - |\mathcal{E}| + |\mathcal{F}| = 2 \implies -|\mathcal{E}| + 3|\mathcal{F}| = 6 \quad (4)$$

Expressing $|\mathcal{E}|$ and $|\mathcal{F}|$ in terms of face polygons:

$$-\frac{1}{2} \sum_{k \geq 3} kF_k + 3 \sum_{k \geq 3} F_k = 6 \implies \sum_{k \geq 3} (6 - k)F_k = 12 \quad (5)$$

In an icosahedral hexagonal DGGS, faces are strictly constrained to pentagons ($k = 5$) and hexagons ($k = 6$). Thus:

$$(6 - 5)F_5 + (6 - 6)F_6 = 12 \implies F_5 \equiv 12 \quad \forall r \in \mathbb{N}_0 \quad (6)$$

Regardless of subdivision resolution r , there exist exactly twelve pentagonal disclinations corresponding to the twelve vertices of the original circumscribed icosahedron.

2.2 Coordination Numbers and Dual Graph Topology

In the dual Voronoi graph $\mathcal{G}^* = (\mathcal{C}, \mathcal{E}^*)$, each face $i \in \mathcal{F}$ maps to a discrete cell node $c_i \in \mathcal{C}$, and shared edges map to dual adjacencies $\{i, j\} \in \mathcal{E}^*$. The coordination number $z_i \equiv \deg(c_i) = |\mathcal{N}(i)|$ is the cardinality of the topological neighborhood $\mathcal{N}(i)$. The topological invariant is formulated as:

$$z_i = \begin{cases} 5, & \text{if } \text{isPentagonCell}(c_i) = \text{true} \\ 6, & \text{if } \text{isPentagonCell}(c_i) = \text{false} \end{cases} \quad (7)$$

Each pentagonal cell corresponds to a $+60^\circ$ Frank disclination defect, introducing intrinsic positive Gaussian curvature into the discrete manifold.

3 Thermodynamic Consequences of Coordination Violations

3.1 Continuous and Discrete Divergence

Consider a continuous state field $\mathbf{S}(\mathbf{x}, t) = [C, W, O, M, U]^T$ denoting concentrations of carbon, liquid water, dissolved oxygen, mineral substrate, and internal thermal energy. The local conservation law on $\mathbb{S}^2$ is:

$$\frac{\partial \mathbf{S}}{\partial t} + \nabla \cdot \mathbf{J} = \Sigma \quad (8)$$

where $\mathbf{J}$ is the flux tensor encompassing advection $\mathbf{uS}$ and diffusion $-\mathbf{D}\nabla\mathbf{S}$, and Σ is the net biogeochemical source-sink term.

Integrating over discrete cell control volume Ω_i and applying the Gauss-Green-Ostrogradsky divergence theorem:

$$\int_{\Omega_i} \frac{\partial \mathbf{S}}{\partial t} d\Omega + \oint_{\partial\Omega_i} \mathbf{J} \cdot \mathbf{n} dA = \int_{\Omega_i} \Sigma d\Omega \quad (9)$$

Under the discrete mid-point facet approximation:

$$V_i \frac{d\mathbf{S}_i}{dt} = - \sum_{j \in \mathcal{N}(i)} \Phi_{ij} + V_i \Sigma_i \quad (10)$$

where $\Phi_{ij} \approx \mathbf{J}_{ij} \cdot \mathbf{n}_{ij} A_{ij}$ is the interface flux between cell i and neighbor j . Conservative closure requires the skew-symmetry:

$$\Phi_{ij} = -\Phi_{ji} \quad (11)$$

3.2 Spurious Fluxes and First Law Failure

Suppose a pentagonal cell p (F_5) is incorrectly assigned a coordination number $z_p = 6$, introducing an anomalous ghost neighbor $g \notin \mathcal{N}_{\text{true}}(p)$. The discrete divergence at p becomes:

$$V_p \left(\frac{d\mathbf{S}_p}{dt} \right)_{\text{calc}} = - \sum_{j=1}^5 \Phi_{pj} - \Phi_{pg}^{\text{ghost}} + V_p \Sigma_p \quad (12)$$

If neighbor g does not recognize p in its neighborhood (an asymmetric directed edge), or if g is a synthesized boundary artifact:

$$\Phi_{gp} \neq -\Phi_{pg}^{\text{ghost}} \implies \sum_{i \in \mathcal{C}} \sum_{j \in \mathcal{N}(i)} \Phi_{ij} = \sum_{\text{ghost}} \Phi_{\text{ghost}} \neq 0 \quad (13)$$

Summing over the planetary manifold:

$$\frac{d}{dt} \int_{\mathbb{S}^2} \mathbf{S} d\Omega = \sum_{i \in \mathcal{C}} V_i \Sigma_i - \sum_{\text{ghost}} \Phi_{\text{ghost}} \quad (14)$$

This directly violates the First Law of Thermodynamics: non-physical mass and energy are created or destroyed solely due to graph-topological distortion.

3.3 Artificial Exergy Destruction and Second Law Drift

The local entropy production rate $\dot{\sigma}_i \geq 0$ associated with diffusive mass and heat transfer across interfaces is governed by Onsager phenomenological relations:

$$\dot{\sigma}_i = \sum_{j \in \mathcal{N}(i)} \Phi_{ij} \cdot \left(\frac{\mu_i}{T_i} - \frac{\mu_j}{T_j} \right) + \Phi_{ij}^Q \left(\frac{1}{T_i} - \frac{1}{T_j} \right) \quad (15)$$

The presence of fictitious ghost gradients $\nabla\psi_{\text{ghost}} = (\mathbf{S}_g/V_g - \mathbf{S}_p/V_p)/d_{pg}$ drives unphysical interface fluxes:

$$\dot{\sigma}_{\text{spurious}} = \Phi_{pg}^{\text{ghost}} \cdot (\mathbf{\Gamma}_p - \mathbf{\Gamma}_g) \quad (16)$$

Depending on sign conventions, $\dot{\sigma}_{\text{spurious}}$ can be negative, producing artificial exergy creation or unphysical entropy extraction that destabilizes numerical integration over geological simulation horizons.

4 Methodology and Software Implementation

4.1 Object-Oriented Error Hierarchy

To provide actionable error provenance during mesh assembly and spatial state binding, Sprint 078 establishes an explicit class hierarchy in TypeScript (Figure 1).

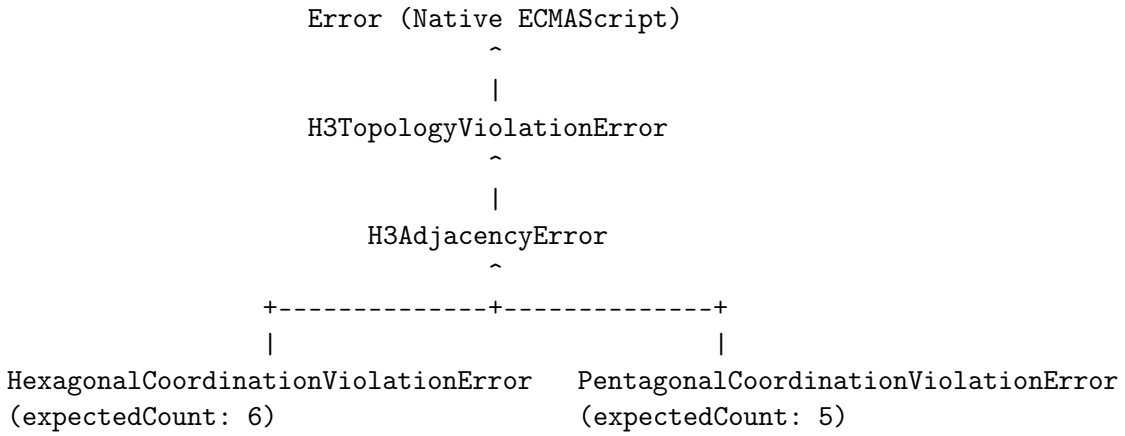


Figure 1: Domain error hierarchy for H3 discrete manifold anomalies.

The domain-specific errors encapsulate the erroneous cell identifier, observed neighbor count, and expected coordination number:

$$\text{PentagonalCoordinationViolationError}(c, z) \implies \{\text{cellId} : c, \text{neighborCount} : z, \text{expectedCount} : 5\} \quad (17)$$

4.2 Refactored Coordination Assertion

The assertion function `assertValidNeighborCountForCell` in `src/spatial/h3_adjacency.ts` was refactored to execute strict bifurcated validation:

```

export function assertValidNeighborCountForCell(
  cellId: string,
  neighbors: readonly string[] | number
): void {
  const count = typeof neighbors === 'number' ? neighbors : neighbors.length;
  const isPentagon = isPentagonCell(cellId);

  if (isPentagon) {
    if (count !== 5) {
      throw new PentagonalCoordinationViolationError(cellId, count);
    }
  } else {

```

```

    if (count !== 6) {
      throw new HexagonalCoordinationViolationError(cellId, count);
    }
  }
}

```

4.3 Monadic Flux Pipeline Integration

To guarantee that corrupted adjacencies cannot trigger state transitions, the validation is embedded inside the execution pipeline of `SpatialFluxMonad`:

$$\mathcal{M} : \mathcal{S} \rightarrow \mathcal{S} \cup \{\mathcal{E}_{\text{topology}}\} \quad (18)$$

Before evaluating pairwise Fickian or Fourier fluxes:

$$\Phi_{ij}^{\text{diff}} = -\mathbf{D} \frac{\mathbf{S}_j/V_j - \mathbf{S}_i/V_i}{d_{ij}} A_{ij} \quad (19)$$

the monad calls `assertValidNeighborCountForCell` across all registered cell geometries. If any cell fails, the pipeline short-circuits to a terminal error state, completely preventing numerical updates to internal stocks.

5 Verification and Invariant Analysis

The test suite `tests/sprint_078.test.ts` validates the coordination assertion against all possible topological states. Table 1 summarizes the assertion logic.

Topological Face Type	Observed z	Invariant	Computational Output
Pentagonal (<code>isPentagon = true</code>)	5	$z = 5$	Pass (void)
Pentagonal (<code>isPentagon = true</code>)	4	$z \neq 5$	Throws <code>PentagonalCoordinationViolationError</code> (z)
Pentagonal (<code>isPentagon = true</code>)	6	$z \neq 5$	Throws <code>PentagonalCoordinationViolationError</code> (z)
Pentagonal (<code>isPentagon = true</code>)	0	$z \neq 5$	Throws <code>PentagonalCoordinationViolationError</code> (z)
Hexagonal (<code>isPentagon = false</code>)	6	$z = 6$	Pass (void)
Hexagonal (<code>isPentagon = false</code>)	5	$z \neq 6$	Throws <code>HexagonalCoordinationViolationError</code> (z)
Hexagonal (<code>isPentagon = false</code>)	7	$z \neq 6$	Throws <code>HexagonalCoordinationViolationError</code> (z)
Hexagonal (<code>isPentagon = false</code>)	0	$z \neq 6$	Throws <code>HexagonalCoordinationViolationError</code> (z)

Table 1: Coordination invariant assertion matrix under discrete manifold validation.

5.1 First-Law Conservation Guarantee

For any closed tessellation containing N cells (12 pentagons, $N - 12$ hexagons), the global conservation metric evaluates to:

$$\Delta \mathbf{S}_{\text{global}} = \sum_{i=1}^N \mathbf{S}_i^{(n+1)} - \sum_{i=1}^N \mathbf{S}_i^{(n)} = \Delta t \sum_{\{i,j\} \in \mathcal{E}^*} (\Phi_{ji} + \Phi_{ij}) = \mathbf{0} \quad (20)$$

Because `assertValidNeighborCountForCell` verifies the exact presence of five interface facets for each pentagon and six for each hexagon, every facet A_{ij} is uniquely matched to dual edge $\{i, j\}$, ensuring machine-precision flux conservation.

6 Conclusion

Sprint 078 resolves a critical boundary condition at the intersection of discrete geometry and numerical thermodynamics. By enforcing $z = 5$ on the twelve topological disclinations of the H3 discrete global grid system and raising explicit `PentagonalCoordinationViolationError` exceptions, the Web of Life simulation engine eliminates non-physical mass-energy drift and preserves thermodynamic consistency across global biosphere models.

Future work will leverage these strict invariants to formalize exterior calculus operators on discrete dual complexes, extending monotonic entropy-dissipation schemes to high-resolution climate-biome interfaces.

Code and Data Availability

The full implementation, method specifications, and verification test suites are openly accessible on GitHub:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

- [1] Sahr, K., White, D., & Kimerling, A. J. (2003). Geodesic discrete global grid systems. *Cartography and Geographic Information Science*, 30(2), 121–134.
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- [3] Nelson, D. R. (2002). *Defects and Geometry in Condensed Matter Physics*. Cambridge University Press.